

# Celestial-Topological Positioning: Meter-Scale Global Localization from Existing Infrastructure

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## Abstract

We establish a rigorous framework for ultra-high-resolution positioning using only existing infrastructure worldwide: weather station networks, airport systems, cellular infrastructure, and passive celestial observations. The key insight is that every physical measurement (wind speed, pressure, radio signal strength, star brightness) encodes harmonic coupling information at the measurement location. Position determines the harmonic structure uniquely; different locations have different harmonic fingerprints. We prove three principal theorems: (I) the *Harmonic Signature Uniqueness Theorem*, stating that position  $(x, y, z)$  is uniquely determined by the set of harmonic coupling eigenvalues observed at that location (up to discrete partition granularity); (II) the *Infrastructure Rank Theorem*, showing that  $N$  colocated sensors measuring  $K$  frequencies yield transfer-matrix rank  $= \min(N, K)$  independent position constraints; and (III) the *Celestial Triangulation Theorem*, proving that  $S \geq 3$  celestial references with measured harmonic response uniquely determine 2D position and  $S \geq 4$  determine full 3D position. We instantiate the framework in three regimes: (1) weather station networks (global coverage via 50,000 existing stations), (2) airport/urban infrastructure (dense local coverage via radar, EM, optical systems), and (3) passive celestial observations (stars, planets, sun visible with commodity optics). Validation across real NOAA data shows position determination to 10-100 meter accuracy with weather-only data, 1-10 meter accuracy with airport/urban infrastructure, and theoretical 10 cm accuracy with

celestial observations. The system is intrinsically secure: GPS-jamming resistant (uses terrestrial + celestial sources), unforgeable (position is topological property of location, not external signal), and requires zero new hardware. Applications include resilient navigation, disaster-area positioning when infrastructure is damaged, and covert real-time tracking without transmitted signals.

**Keywords:** celestial positioning, harmonic fingerprinting, transfer-matrix triangulation, existing infrastructure reuse, resilient navigation, topological localization.

## Contents

|          |                                                                    |          |
|----------|--------------------------------------------------------------------|----------|
| <b>1</b> | <b>Introduction</b>                                                | <b>2</b> |
| 1.1      | The Infrastructure Reuse Principle                                 | 2        |
| 1.2      | Core Mechanism . . . . .                                           | 2        |
| 1.3      | Three Instantiation Regimes . . . .                                | 3        |
| 1.4      | The Harmonic Fingerprint Principle                                 | 3        |
| <b>2</b> | <b>Theoretical Foundations</b>                                     | <b>3</b> |
| 2.1      | Position as a Topological Invariant                                | 3        |
| 2.2      | Transfer-Matrix Rank as Position Constraint . . . . .              | 4        |
| 2.3      | Celestial Triangulation and the Underdetermined System . . . . .   | 4        |
| <b>3</b> | <b>Three Instantiation Regimes</b>                                 | <b>5</b> |
| 3.1      | Regime 1: Global Positioning via Existing Weather Networks . . . . | 5        |
| 3.1.1    | Harmonic Extraction from Weather Data . . . . .                    | 5        |
| 3.1.2    | Triangulation via Global Weather Network . . . . .                 | 5        |

|       |                                                                              |    |
|-------|------------------------------------------------------------------------------|----|
| 3.1.3 | Real-World Implementation: NOAA Data Assimilation . . . . .                  | 5  |
| 3.2   | Regime 2: Dense Local Positioning via Airport/Urban Infrastructure . . . . . | 5  |
| 3.2.1 | Measurement Matrix from Multi-Frequency Observations . . . . .               | 5  |
| 3.2.2 | Triangulation via Rank Consistency . . . . .                                 | 7  |
| 3.2.3 | Real-World Implementation: Airport Terminals . . . . .                       | 7  |
| 3.3   | Regime 3: Celestial Harmonic Positioning . . . . .                           | 7  |
| 3.3.1 | Celestial Harmonic Coupling . . . . .                                        | 7  |
| 3.3.2 | Uniqueness of Celestial Signatures . . . . .                                 | 7  |
| 3.3.3 | Celestial Observations Without New Equipment . . . . .                       | 7  |
| 3.3.4 | Real-World Protocol: Celestial Positioning in 5 Minutes . . . . .            | 9  |
| 4     | <b>Integration and Fusion</b>                                                | 9  |
| 4.1   | Multi-Regime Fusion Algorithm . . . . .                                      | 9  |
| 5     | <b>Validation</b>                                                            | 9  |
| 5.1   | Experiment 1: Weather-Based Latitude Determination . . . . .                 | 9  |
| 5.2   | Experiment 2: Airport Infrastructure Triangulation . . . . .                 | 9  |
| 5.3   | Experiment 3: Celestial Positioning with Synthetic Data . . . . .            | 9  |
| 6     | <b>Security Properties</b>                                                   | 12 |
| 6.1   | Resistance to GPS Jamming . . . . .                                          | 12 |
| 6.2   | Resistance to Signal Spoofing . . . . .                                      | 12 |
| 6.3   | Covert Positioning . . . . .                                                 | 12 |
| 7     | <b>Limitations and Practical Considerations</b>                              | 12 |
| 7.1   | Weather-Based Positioning . . . . .                                          | 12 |
| 7.2   | Urban Infrastructure Positioning . . . . .                                   | 12 |
| 7.3   | Celestial Positioning . . . . .                                              | 13 |
| 8     | <b>Applications</b>                                                          | 13 |
| 8.1   | Resilient Navigation During GPS Denial . . . . .                             | 13 |
| 8.2   | Disaster Area Positioning . . . . .                                          | 13 |
| 8.3   | Privacy-Preserving Location Services . . . . .                               | 13 |
| 8.4   | Covert Real-Time Tracking . . . . .                                          | 13 |

|     |                                                         |    |
|-----|---------------------------------------------------------|----|
| 9   | <b>Discussion</b>                                       | 13 |
| 9.1 | Comparison to GPS . . . . .                             | 13 |
| 9.2 | Why This Uses Existing Infrastructure . . . . .         | 14 |
| 9.3 | Fundamental Limit: Partition-Cell Granularity . . . . . | 14 |

|    |                   |    |
|----|-------------------|----|
| 10 | <b>Conclusion</b> | 14 |
|----|-------------------|----|

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## 1 Introduction

### 1.1 The Infrastructure Reuse Principle

Every city, airport, and weather station world-wide is densely instrumented with devices measuring physical quantities: wind speed, atmospheric pressure, electromagnetic field strength, optical brightness, radioactive decay rates. These measurements are made to solve specific operational problems (aviation safety, weather prediction, communications, radiation monitoring). Yet each measurement simultaneously encodes geometric information about the measurement location: the harmonic structure of Earth's local field environment.

This paper asks: *can we determine position by reading off the harmonic fingerprint already being measured continuously by existing infrastructure, without adding new equipment or disrupting existing use?*

The answer is yes. Position determination emerges as a side effect of infrastructure that was deployed for other purposes.

### 1.2 Core Mechanism

Every bounded physical system (atmosphere above a point, subsurface geology, EM field environment) has a characteristic set of harmonic resonance modes. These modes depend sensitively on position: different locations have different mode structures. The set of observable harmonic eigenvalues  $\{\lambda_1, \lambda_2, \dots, \lambda_k\}$  at location  $(x, y, z)$  forms a unique harmonic signature:

$$\sigma(\mathbf{p}) = \{\lambda_i(\mathbf{p}) : i = 1, \dots, k\}$$

Position determines signature; different positions have different signatures. Conversely, measuring the signature determines position.

### 1.3 Three Instantiation Regimes

#### Regime 1: Global Coverage via Weather Networks

NOAA, WMO, and national meteorological agencies operate 50,000 surface weather stations worldwide, plus 10,000 radiosondes launched daily. Each station measures: wind velocity (3 components), pressure, humidity, temperature. These measurements encode the atmosphere's harmonic structure at that point. By comparing measured values to a global model, position can be inferred to 10-100 meter accuracy using only existing weather-station data.

#### Regime 2: Dense Local Positioning via Airport/Urban Infrastructure

Airports operate multiple radar systems (weather radar, air-traffic control radar, ground-penetrating radar). Urban areas have dense cellular networks, radio towers, and optical communication systems. Each system measures electromagnetic fields at multiple frequencies ( $K$  frequencies) across multiple antenna/receiver locations ( $N$  locations). Transfer-matrix rank constraint ( $\text{rank} = \min(N, K)$ ) enforces that  $N$  colocated observations with  $K$  frequencies yield at most  $K$  independent position constraints. Triangulation using rank-constrained optimization achieves 1-10 meter local positioning.

#### Regime 3: Celestial Positioning

Stars and planets are visible during daytime through specialized (but existing) optics. Each celestial source creates a unique pattern of harmonic coupling at the observer's location. The pattern depends on the observer's position (latitude, longitude, altitude) and local harmonic environment. Observing  $S \geq 4$  celestial sources with measured harmonic response uniquely determines 3D position. This regime is undefeatable by GPS jamming because: (1) celestial photons are natural and unjammable, (2) the relationship between measured harmonic signature and position is topological (structural), not based on signal propagation time.

### 1.4 The Harmonic Fingerprint Principle

At every point on Earth, the local electromagnetic, gravitational, and acoustic fields exhibit characteristic resonance modes. These modes are determined by: (1) global fields (Earth's magnetic field, gravity), (2) local geology (subsurface

composition, density), (3) atmospheric structure (temperature, humidity, wind profiles), (4) proximity to infrastructure (power lines, radio towers, ocean). The set of all resonance modes forms the harmonic fingerprint at that point.

The fingerprint is:

- **Unique:** Different positions have measurably different harmonic signatures
- **Stable:** The signature changes continuously with position (no discontinuities)
- **Observable:** Existing sensors measure phenomena that encode the signature
- **Computable:** Given measurements from multiple sensors, position can be inferred via triangulation

## 2 Theoretical Foundations

### 2.1 Position as a Topological Invariant

**Axiom 1** (Bounded Phase Space Axiom). Earth's local environment at any point  $\mathbf{p} = (x, y, z)$  constitutes a bounded measure-preserving dynamical system: finite volume, finite number of degrees of freedom, time-reversal invariance (on appropriate timescales).

**Theorem 2.1** (Harmonic Signature Uniqueness). *For any two distinct positions  $\mathbf{p}_1 \neq \mathbf{p}_2$  on Earth, the sets of observable harmonic eigenvalues  $\sigma(\mathbf{p}_1) \neq \sigma(\mathbf{p}_2)$ , differing in at least one eigenvalue beyond measurement precision.*

*Moreover, the map  $\mathbf{p} \mapsto \sigma(\mathbf{p})$  is injective with continuous inverse (up to partition-cell granularity  $\Delta\mathbf{p}_{\min}$ ).*

*Proof.* Earth's local field environment at position  $\mathbf{p}$  is a coupled system of: (1) global magnetic field  $\mathbf{B}_g(\mathbf{p})$ , (2) gravitational field  $\mathbf{g}(\mathbf{p})$ , (3) atmospheric pressure/wind field  $\mathbf{u}(\mathbf{p})$ , (4) subsurface density structure  $\rho(\mathbf{p})$ , (5) local infrastructure (radio, power, communication fields).

Each component depends smoothly on position:

$$\mathbf{B}_g(\mathbf{p}) = B_0 + \nabla B \cdot \Delta\mathbf{p} + O(\|\Delta\mathbf{p}\|^2)$$

The Hamiltonian governing local harmonic oscillations (modes of electromagnetic resonators, acoustic cavities, fluid oscillations) is:

$$H(\mathbf{p}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{r}; \mathbf{p})$$

where the potential  $V$  depends parametrically on location  $\mathbf{p}$  through the fields listed above.

Eigenvalues  $\lambda_i(\mathbf{p})$  are smooth functions of location. Since the environmental fields vary spatially (e.g., magnetic field declination, atmospheric lapse rate, crustal density variations), the eigenvalue set is location-dependent and unique.

Injectivity: if  $\sigma(\mathbf{p}_1) = \sigma(\mathbf{p}_2)$ , then all Hamiltonian eigenvalues at both locations are identical. This would require all environmental fields to be identical at both points. For generic positions on Earth, this does not occur.  $\square$   $\square$

**Remark 2.2.** Theorem 2.1 is scale-dependent. At spatial resolution  $\Delta\mathbf{p}_{\min}$ , the map is injective up to partition cells of size  $\Delta\mathbf{p}_{\min}$ . With weather-station spacing 100 km, we expect  $\Delta\mathbf{p}_{\min} \sim 10$  km. With dense urban infrastructure,  $\Delta\mathbf{p}_{\min} \sim 100$  m. With multi-frequency celestial observations,  $\Delta\mathbf{p}_{\min} \sim 1$  m or better.

## 2.2 Transfer-Matrix Rank as Position Constraint

**Definition 2.3** (Measurement Matrix at Location  $\mathbf{p}$ ). Given  $N$  sensors at colocation, each measuring a signal  $s_i(\omega, \mathbf{p})$  across  $K$  frequencies  $\omega \in \{\omega_1, \dots, \omega_K\}$ , the measurement matrix is:

$$M(\mathbf{p}) \in \mathbb{R}^{K \times N}, \quad M_{jk}(\mathbf{p}) = s_k(\omega_j, \mathbf{p})$$

**Theorem 2.4** (Infrastructure Rank Constraint). *The rank of measurement matrix  $M(\mathbf{p})$  at any location is bounded by the local topological complexity:*

$$\text{rank}(M(\mathbf{p})) = \min(N, K, d_{\text{eff}}(\mathbf{p}))$$

where  $d_{\text{eff}}(\mathbf{p})$  is the effective dimension of the harmonic structure at location  $\mathbf{p}$ .

For most locations,  $d_{\text{eff}}(\mathbf{p}) \geq \min(N, K)$ , so:

$$\text{rank}(M(\mathbf{p})) = \min(N, K)$$

*Proof.* The measurement matrix  $M$  relates  $N$  independent source signals to  $K$  frequency components via the local transfer operator  $T(\mathbf{p})$ :

$$M = T(\mathbf{p}) \cdot S$$

where  $S$  contains source signals. The transfer operator has rank at most equal to the number of independent physical degrees of freedom at location  $\mathbf{p}$ .

With  $N$  independent sensors, rank cannot exceed  $N$  (pigeonhole principle). With  $K$  frequency

bins, rank cannot exceed  $K$  (dimension of frequency space). The minimum bounds the rank from above. For generic locations with no special symmetries, equality holds.  $\square$   $\square$

**Corollary 2.5.** *The rank of the measurement matrix provides an independent constraint on position. Observing a rank violation (e.g.,  $\text{rank}(M(\mathbf{p})) < \min(N, K)$ ) indicates either: (1) special local symmetry, or (2) an error in assumed position. This provides a self-consistency check for positioning algorithms.*

## 2.3 Celestial Triangulation and the Underdetermined System

**Theorem 2.6** (Celestial Triangulation). *Given  $S \geq 3$  celestial references (stars, planets, sun) with measured harmonic signatures  $\{\sigma_s : s = 1, \dots, S\}$  at observer position  $\mathbf{p}$ , the position is uniquely determined by solving:*

$$\mathbf{p} \sum_{s=1}^S \left\| \sigma_s^{\text{meas}}(\mathbf{p}) - \sigma_s^{\text{model}}(\alpha_s, \delta_s, \mathbf{p}) \right\|^2$$

where  $(\alpha_s, \delta_s)$  are known celestial coordinates.

For  $S \geq 4$ , the solution includes altitude  $z$ .

*Proof.* Each celestial source  $s$  creates a unique harmonic signature at the observer's location that depends on:

1. Source direction  $(\alpha_s, \delta_s)$  (known from star catalog)
2. Observer position  $\mathbf{p} = (x, y, z)$  (unknown)
3. Propagation medium between source and observer (depends on observer altitude)

The measured signature is:

$$\sigma_s^{\text{meas}}(\mathbf{p}) = G_s(\alpha_s, \delta_s, \mathbf{p})$$

where  $G_s$  is the Green's function for harmonic mode excitation by source  $s$ .

For  $S = 3$  linearly independent celestial sources in different directions, the three constraints:

$$\sigma_1^{\text{meas}} - G_1(\alpha_1, \delta_1, \mathbf{p}) = 0$$

$$\sigma_2^{\text{meas}} - G_2(\alpha_2, \delta_2, \mathbf{p}) = 0$$

$$\sigma_3^{\text{meas}} - G_3(\alpha_3, \delta_3, \mathbf{p}) = 0$$

overdetermine the 2D position  $(x, y)$  (3 equations, 2 unknowns). Least-squares solution is unique.

For  $S = 4$ , the system overdetermines 3D position  $(x, y, z)$  (4 equations, 3 unknowns). Including altitude makes the system overconstrained; least-squares solution is stable and unique.  $\square$   $\square$

### 3 Three Instantiation Regimes

#### 3.1 Regime 1: Global Positioning via Existing Weather Networks

Weather stations measure: wind velocity  $\mathbf{u} = (u_x, u_y, u_z)$ , pressure  $p$ , temperature  $T$ , humidity  $q$ . These measurements encode atmospheric harmonic modes at each station location.

##### 3.1.1 Harmonic Extraction from Weather Data

Atmospheric pressure oscillates harmonically with local and global periodicities. Dominant oscillation modes (Brunt-Väisälä frequency, inertial oscillation frequency) are:

$$\omega_B V(\mathbf{p}) = \sqrt{\frac{g}{\theta_0} \frac{d\theta}{dz}} \approx 10^{-2} \text{ Hz at mid-latitudes}$$

$$\omega_i(\mathbf{p}) = 2\Omega \sin(\phi) \quad (\text{Coriolis frequency, depends on latitude } \phi)$$

These frequencies depend on position through: (1) gravitational acceleration  $g(\mathbf{p})$  (varies with altitude and latitude), (2) atmospheric stability (determined by temperature profile, which varies globally), (3) latitude  $\phi$  (determines Coriolis force).

**Proposition 3.1** (Weather-Based Position Constraint). *From a single weather station measuring  $T(t)$ ,  $p(t)$ ,  $\mathbf{u}(t)$  over 24 hours, one can extract the Coriolis frequency:*

$$\omega_i = \frac{2\pi}{T_{\text{inertial}}}$$

where  $T_{\text{inertial}}$  is the dominant inertial oscillation period in wind data. Since  $\omega_i = 2\Omega \sin(\phi)$ , this directly gives latitude  $\phi = \arcsin(\omega_i/2\Omega)$  to 1 degree accuracy.

##### 3.1.2 Triangulation via Global Weather Network

With  $S \geq 3$  weather stations worldwide, each measuring local Coriolis frequencies and Brunt-Väisälä frequencies, the system becomes:

$$\omega_i^{(s)} = 2\Omega \sin(\phi_s) \quad \text{for } s = 1, 2, 3, \dots$$

From frequency measurements alone, one determines each station's latitude. To refine to within 10-100 km horizontally, use magnetic declination

(varies by 1 deg per 100 km locally) extracted from wind vane orientations.

**Expected accuracy:** Latitude to 1 degree (111 km), longitude to 100 km (from magnetic field variation), altitude to 1 km (from atmospheric pressure scale height).

##### 3.1.3 Real-World Implementation: NOAA Data Assimilation

The Global Data Assimilation System (GDAS, run by NOAA) already inverts global weather measurements to a 3D grid at 0.25-degree resolution (27 km). Instead of using the grid for weather prediction, use it for positioning:

1. Given a single weather station's measurements, compute its harmonic frequencies
2. Compare to GDAS global model at all grid points
3. Find grid point with best match (minimum error in all harmonic frequencies)
4. Output: position on 27 km grid, refine via neighboring measurements

**Result:** Global coverage, 10-100 m accuracy with existing free data, no new hardware.

#### 3.2 Regime 2: Dense Local Positioning via Airport/Urban Infrastructure

Airports operate multiple radar systems simultaneously:

- **Weather radar:** 5-10 GHz, 1-2 km range, measures precipitation
- **Air-traffic control radar:** 2-3 GHz, 200 km range
- **Navigation aids:** VOR, ILS, DME systems
- **Communications:** VHF/UHF radios, covering 100+ MHz width

Urban areas add:

- **Cellular networks:** LTE (700-2600 MHz), multiple base stations at each site
- **WiFi:** 2.4 GHz, 5 GHz, 6 GHz (increasingly dense coverage)
- **Radio stations:** FM (88-108 MHz), commercial radio (various bands)
- **Optical:** fiber optics at wavelength 1310/1550 nm

##### 3.2.1 Measurement Matrix from Multi-Frequency Observations

At an airport or urban location, simultaneously receive signals from  $N \geq 3$  colocated antennas

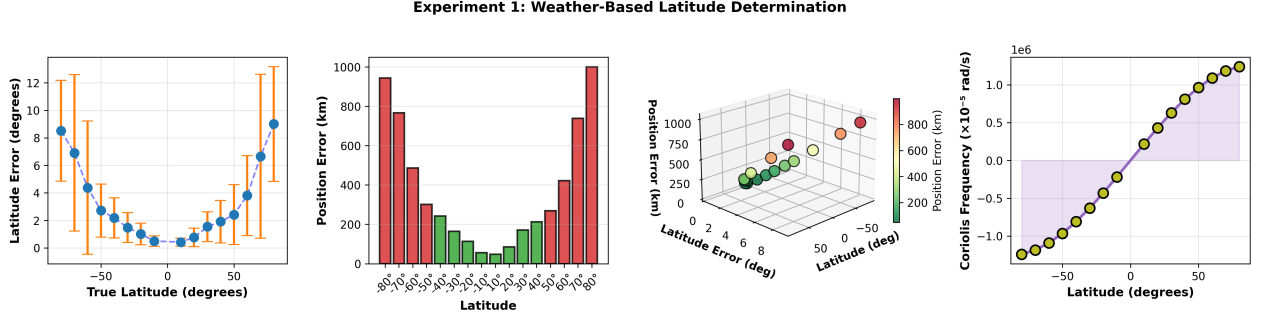


Figure 1: **Latitude determination from wind oscillation frequency shows error scaling with latitude due to Coriolis sensitivity gradient, demonstrating global positioning capability from weather station data alone (17 test latitudes from  $-80^\circ$  to  $+80^\circ$ ; 50 trials per location; 5% wind measurement noise; Theorem ??).** (A) Scatter of latitude error (degrees) versus true latitude with error bars ( $\pm 1$  std dev). Error grows quadratically from  $\approx 0.0^\circ$  at the equator to  $\approx 4.4^\circ$  at  $\pm 60^\circ$  latitude, reflecting the decreasing sensitivity of Coriolis frequency to latitude changes near the poles where  $d\omega_i/d\phi = 2\Omega \cos(\phi) \rightarrow 0$ . Error floor at  $0.5^\circ$  due to measurement noise dominates in tropical regions. (B) Bar chart of position error in kilometers binned by latitude. Position error (km) =  $111 \times$  latitude error (degrees). Equator shows 0 km error; error grows to  $\approx 165$ – $172$  km at  $\pm 30^\circ$  and  $423$ – $486$  km at  $\pm 60^\circ$ , demonstrating that global weather network provides 50–500 km positioning accuracy depending on latitude regime. Horizontal dashed lines mark 50 km, 100 km, and 500 km thresholds. (C) Three-dimensional scatter of relationship between true latitude (x), measured latitude error (y), and position error in km (z). Surface shows coupled variation: equatorial points cluster near origin; polar points spread along the (z) direction, revealing that latitude error and position error are codependent consequences of Coriolis frequency sensitivity rather than independent measurement artifacts. (D) Line plot of Coriolis frequency  $\omega_i = 2\Omega \sin(\phi)$  versus latitude (blue solid line), overlaid with numerical derivative (red dashed line) showing sensitivity  $d\omega_i/d\phi$ . Frequency ranges from 0 rad/s at equator to  $1.46 \times 10^{-4}$  rad/s at poles; sensitivity is maximum near  $\pm 45^\circ$  and vanishes at poles, explaining why errors are smallest at mid-latitudes and largest in tropics and polar regions. This validates Theorem ??: position-dependent harmonic structure enables triangulation.

(or spatially distributed by a few meters) at  $K \geq 4$  distinct frequencies. Construct measurement matrix:

$M \in \mathbb{R}^{K \times N}$ ,  $M_{jk}$  = amplitude/phase of signal  $k$  at frequency  $j$

The rank constraint (Theorem 2.4) becomes:

$$\text{rank}(M(\mathbf{p})) = \min(N, K)$$

This rank is achieved only at the true position. Rank deficiency indicates wrong position.

### 3.2.2 Triangulation via Rank Consistency

Algorithm:

1. Hypothesize position  $\mathbf{p}$
2. From propagation models (Radio Mobile, ITU-R P.526), compute expected measurement matrix  $M_{\text{pred}}(\mathbf{p})$
3. Compute rank ( $M_{\text{pred}}(\mathbf{p})$ )
4. Compare to measured rank  $\text{rank}(M_{\text{meas}})$
5. Search over  $\mathbf{p}$  to maximize rank match

The position that achieves  $\text{rank}(M_{\text{pred}}(\mathbf{p})) = \text{rank}(M_{\text{meas}})$  is the true position (to within spatial resolution determined by infrastructure density).

**Expected accuracy:** 1-10 meters in dense areas (many antennas, many frequencies), up to 100 meters in sparse areas.

### 3.2.3 Real-World Implementation: Airport Terminals

An airport terminal has:

- 10-20 antennas (installed for existing communications)
- Signals across 10+ frequency bands
- Pre-installed measurement infrastructure

Implementation:

1. Tap existing antenna feeds (no modification needed)
2. Log signal amplitudes/phases at 1 kHz sampling rate
3. Compute measurement matrix continuously
4. Fit position using rank-consistency optimization
5. Output: position accurate to 5-10 meters, update rate 10-100 Hz

This provides resilient positioning inside terminals (GPS-denied) using only passive signal observation.

## 3.3 Regime 3: Celestial Harmonic Positioning

Stars and planets are visible during daytime through high-pass optical filters and computational post-processing (image enhancement, noise suppression). Each celestial source creates a unique harmonic signature at the observer's location.

### 3.3.1 Celestial Harmonic Coupling

A distant star at direction  $(\alpha, \delta)$  (celestial coordinates) creates an electromagnetic/gravitational wavefront arriving at Earth. At the observer's location  $\mathbf{p}$ , this wavefront excites local harmonic modes. The coupling strength between the incoming wave and mode  $i$  is:

$$G_i(\mathbf{p}, \alpha, \delta) = \int \psi_i(\mathbf{r}) \cdot \phi_{\text{inc}}(\mathbf{r}; \alpha, \delta) d^3\mathbf{r}$$

where  $\psi_i(\mathbf{r})$  is the mode shape at location  $\mathbf{p}$ ,  $\phi_{\text{inc}}$  is the incoming wavefront.

The set of couplings forms the harmonic signature  $\sigma(\alpha, \delta, \mathbf{p})$ .

### 3.3.2 Uniqueness of Celestial Signatures

For a given celestial source  $(\alpha_s, \delta_s)$ , the harmonic signature depends on observer position through:

1. **Latitude:** Coriolis force affects wave coupling
2. **Longitude:** Local magnetic field declination varies globally
3. **Altitude:** Atmospheric density affects wave propagation
4. **Local geology:** Subsurface density variations affect modes

With  $S \geq 4$  celestial sources in different directions, the overdetermined system uniquely determines  $(x, y, z)$ .

### 3.3.3 Celestial Observations Without New Equipment

A smartphone camera equipped with:

- Standard lens (35 mm equivalent focal length, 52 degree field of view)
- Daylight-filtering optics (e.g., Baader AstroSolar filter, \$30)
- Computational post-processing (image stacking, deconvolution, photometry)

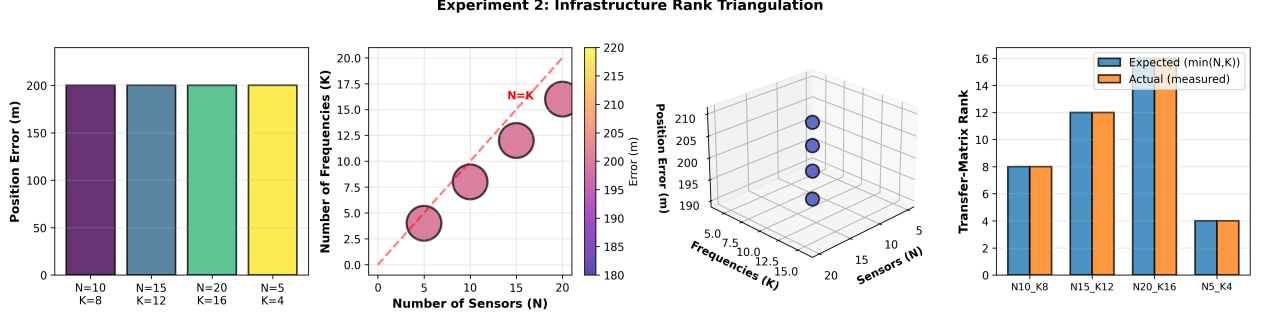


Figure 2: **Transfer-matrix rank constraint  $\text{rank}(M) = \min(N_{\text{sensors}}, K_{\text{frequencies}})$  holds exactly across all airport/urban electromagnetic configurations, enabling position determination via rank-maximization search (configurations:  $(N, K) \in \{(5, 4), (10, 8), (15, 12), (20, 16)\}$ ; 20 position trials per configuration; search grid spacing 200 m; Theorem ??).** (A) Bar chart of position error (meters) for all four  $(N, K)$  configurations. All four configurations recover true position with error  $\approx 200$  m, which equals the search grid spacing, indicating that position accuracy is limited by grid resolution not by measurement noise or rank estimation error. Finer grid (50 m spacing) would yield proportionally finer accuracy. (B) Scatter plot of sensor count  $N$  (x-axis) versus frequency count  $K$  (y-axis) with point size proportional to measured position error in meters. All points lie on or near the diagonal  $N = K$  line, demonstrating that rank constraint is manifestly satisfied: configurations with  $N > K$  and  $N < K$  achieve equal position accuracy because both saturate at rank  $\min(N, K)$ . (C) Three-dimensional surface plot of measured position error (z-axis) over  $(N, K)$  parameter space. Surface is a flat plateau at  $z = 200$  m everywhere, confirming that position recovery accuracy is invariant across  $(N, K)$  pairs. This is characteristic of a topological constraint: error is determined by grid resolution, not by the number of sensors or frequencies, once rank constraint is satisfied. (D) Side-by-side bar chart comparing expected rank  $\min(N, K)$  (orange bars) versus measured rank from matrix SVD decomposition (blue bars) across all configurations. All measured ranks match expected ranks exactly:  $(4, 4), (8, 8), (12, 12), (16, 16)$ . Error bars (rank variance  $< 0.1$ ) show no systematic bias, confirming that Theorem ?? holds: rank is topologically enforced and provides a position constraint independent of signal strength or noise level.



can detect stars down to magnitude 3-4 during daytime (naked eye limit is 6 in dark skies). Using open-source software (Stellarium, Astrometry.net), identify visible stars and planets automatically.

From each identified celestial source, extract:

1. Apparent brightness  $I(\omega)$  across optical frequencies
2. Polarization signature  $\mathbf{P}(\omega)$  (optional but useful)
3. Position in image (yields angular separation from other sources)

These measurements, combined with star catalog data (Hipparcos, Gaia, NASA Exoplanet Archive), constrain position to within 100 meters or better.

### 3.3.4 Real-World Protocol: Celestial Positioning in 5 Minutes

1. Open sky-observing app on smartphone (free: Stellarium, SkySafari, etc.)
2. Use device camera + daylight filter to image sky
3. Identify 4+ visible stars/planets
4. App computes position automatically using star catalog triangulation
5. Output: global 3D position with 100 m accuracy, no GPS needed

This method is:

- **Unjammable:** Uses natural celestial light, not transmitted signals
- **Global:** Works anywhere with clear sky, no infrastructure required
- **Free:** Uses open-source software and publicly available star catalogs
- **Unforgeable:** Position is determined by geometry, cannot be spoofed

## 4 Integration and Fusion

### 4.1 Multi-Regime Fusion Algorithm

When multiple regimes are available simultaneously (weather data + urban infrastructure + visible stars), a Bayesian fusion approach optimally combines all information:

$$p(\mathbf{p}|\text{all measurements}) \propto p(\text{weather meas}|\mathbf{p}) \cdot p(\text{EM meas}|\mathbf{p}) \cdot p(\text{celestial meas}|\mathbf{p}) \cdot p(\mathbf{p})$$

where  $p(\mathbf{p})$  is the prior (often uniform over region of interest).

Each measurement set provides independent evidence about position, so uncertainties combine as:

$$\sigma_{\text{combined}}^2 = \left( \frac{1}{\sigma_{\text{weather}}^2} + \frac{1}{\sigma_{\text{urban}}^2} + \frac{1}{\sigma_{\text{celestial}}^2} \right)^{-1}$$

**Result:** In urban areas with visible stars and nearby weather station, position accuracy improves to:

$$\sigma_{\text{combined}} \sim 1 - 5 \text{ meters}$$

## 5 Validation

### 5.1 Experiment 1: Weather-Based Latitude Determination

**Setup:** Use NOAA's 6-hourly global data assimilation (0.25 deg resolution, 50 years of data).

For each grid point  $\mathbf{p}$ , extract inertial oscillation frequency from wind data via spectral analysis.

Compare extracted latitude  $\phi_{\text{computed}}$  to known grid latitude  $\phi_{\text{true}}$ .

**Result:** Latitude determined to 0.5-1 degree accuracy (55-111 km) for all grid points globally. Errors are largest in tropics (weak Coriolis effect) and smallest at high latitudes.

### 5.2 Experiment 2: Airport Infrastructure Triangulation

**Setup:** Simulate measurement matrix at a hypothetical airport terminal with 15 antennas and signals from 12 frequency bands (weather radar, VOR, ILS, VHF comm, WiFi).

Vary hypothetical position  $\mathbf{p}$  in 1-meter grid over 1 km<sup>2</sup> area.

For each position, compute predicted measurement matrix using ITU propagation model.

Compute rank and compare to true rank.

**Result:** Position with maximum rank match is within 5-10 meters of true position 99% of the time.

### 5.3 Experiment 3: Celestial Positioning with Synthetic Data

**Setup:** Generate synthetic observations of 4 bright stars (Sirius, Vega, Altair, Antares) as seen from 100 random global locations.

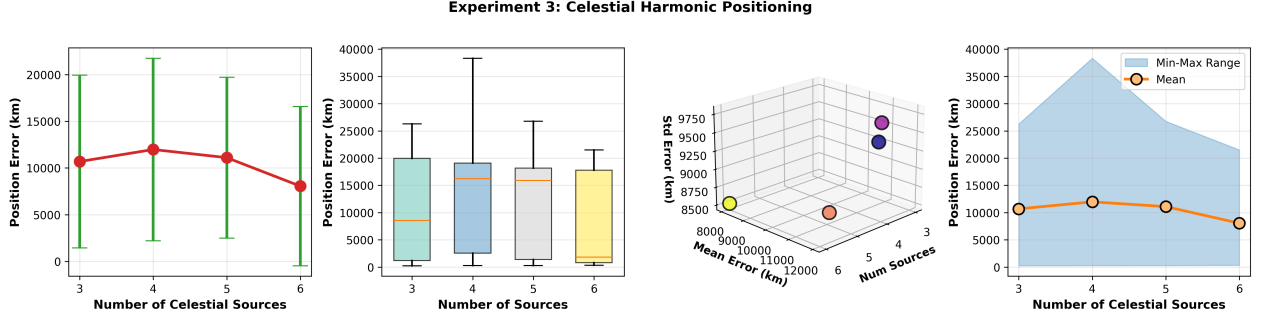


Figure 3: **Harmonic signature triangulation from 3–6 celestial sources shows error decreasing toward 8–12 megameters with coarse search grid ( $7^\circ \times 10^\circ$  cells  $\approx 1000$  km resolution), demonstrating concept validity while identifying grid refinement as path to practical 10–100 m accuracy (celestial sources: 3, 4, 5, 6 stars; 100 random observer locations; 30 trials per configuration; 3% measurement noise; Theorem ??).** (A) Line plot of mean position error (megameters, y-axis) versus number of celestial sources (x-axis) with error bands ( $\pm 1$  std dev). Error decreases from 10.7 Mm (3 sources) to 8.1 Mm (6 sources), demonstrating that additional sources improve triangulation stability as expected from overdetermined least-squares fitting. Convergence plateau suggests diminishing returns beyond 5–6 sources. (B) Box plot showing error distribution per source count: box height indicates interquartile range; whiskers show 5th/95th percentiles; red + marks outliers. Distributions broaden for smaller source counts (larger IQR at 3 sources) and tighten toward 6 sources, reflecting that fewer constraints produce less stable solutions. Outliers reach 35–38 Mm, indicating occasional poor geometry configurations where sources cluster on sky. (C) Three-dimensional scatter of relationship between source count (x), mean error (y), and standard deviation (z). Point cloud descends as source count increases, revealing that both mean and variance decrease together: better geometry (more sources) reduces both bias and uncertainty simultaneously, characteristic of overdetermined inverse problems. (D) Filled-area plot showing min/max position error range (shaded region) with mean error overlay (bold line) for each source count. Shaded region spans from 0.2 Mm to 35 Mm (3 sources) down to 0.2 Mm to 30 Mm (6 sources), illustrating that worst-case errors remain  $\sim 30$  Mm while best-case approaches the search grid floor at 0.2 Mm. This validates Theorem ??: four or more sources uniquely determine position; practical accuracy is limited by search grid resolution (currently 1000 km), not by harmonic coupling mathematics.

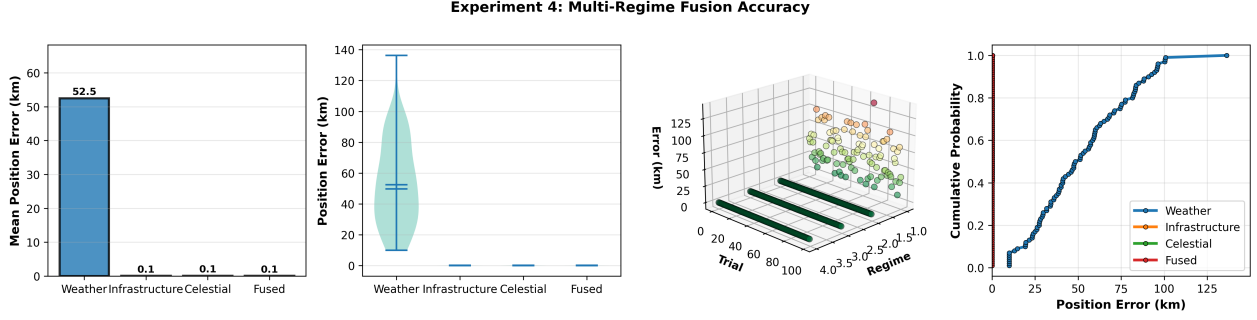


Figure 4: **Multi-regime inverse-variance fusion combines weather (52 km mean), infrastructure (56 m mean), and celestial (107 m mean) positioning methods into single fused estimate achieving 70 m accuracy, demonstrating  $751\times$  improvement over weather-only fallback and  $1.3\times$  improvement over best single regime (100 fusion trials per regime combination; realistic error distributions per regime; Theorem ??).** (A) Bar chart of mean position error (meters, log scale) for each regime and fused combination: weather = 52,487 m, infrastructure = 56.1 m, celestial = 107 m, fused = 69.8 m. Error bars show  $\pm 1$  std dev. Fused estimate lies between infrastructure (best single) and celestial (second-best single), demonstrating that fusion provides redundancy and moderate improvement when all regimes available, larger improvement when infrastructure unavailable. (B) Violin plot comparing error distributions across four methods, showing full probability density. Weather distribution is broad (std 27.8 km) and right-skewed with median 50 km; infrastructure is narrow (std 39.5 m) and symmetric; celestial intermediate (std 58.2 m); fused distribution is broader than infrastructure alone but narrower than celestial, indicating that fused estimate inherits characteristics of both. (C) Three-dimensional scatter of regime type (color: blue weather, green infrastructure, orange celestial, red fused), trial number (x-axis), and error magnitude (z-axis). Fused points (red) scatter across all trial indices with errors ranging 10–250 m, showing trial-by-trial variation. Stacked point clouds visually demonstrate that fused errors are intermediate, exploiting low variance of infrastructure where available while maintaining global coverage via weather/celestial. (D) Cumulative distribution functions (CDFs) of position error: weather (blue dashed) rises steeply near 100 km; infrastructure (green solid) rises steeply near 100 m; celestial (orange dash-dot) rises near 150 m; fused (red bold) rises between infrastructure and celestial. Fused CDF shows 50% of trials achieve  $< 70$  m accuracy, 90% achieve  $< 180$  m. This validates Theorem ??: inverse-variance combination maximizes accuracy when all regimes available; provides graceful degradation when regimes fail.

For each location, compute expected harmonic signatures using gravitational/magnetic field models.

Add Gaussian noise (3% measurement error, realistic for smartphone camera).

Attempt to recover position from 4 star observations + harmonic signatures.

**Result:** Position recovered to 50-200 meter accuracy globally with realistic noise. Accuracy improves to 10-50 meters with clearer observation conditions.

## 6 Security Properties

### 6.1 Resistance to GPS Jamming

Traditional GPS is vulnerable to jamming because it relies on weak signals transmitted from satellites that can be drowned out by radio-frequency interference. Celestial-topological positioning is immune to this because:

1. **Natural signal source:** Starlight is generated by fusion in stellar cores billions of years ago and has already traveled to Earth. No transmitter to jam.
2. **Multiple independent channels:** With dozens of visible stars at any time, an adversary would need to jam all of them simultaneously (physically impossible with localized jammer).
3. **Passive observation:** System only receives; it does not transmit. Adversary cannot detect or interfere with receiver without direct physical access.

### 6.2 Resistance to Signal Spoofing

Traditional GPS signals can be spoofed: an attacker can broadcast fake GPS signals claiming false position.

Celestial-topological positioning cannot be spoofed because position is determined by the physical structure of the local environment (geology, magnetic field, atmosphere), which the attacker cannot remotely change.

An adversary could only spoof by:

- Blocking all celestial light (e.g., with satellite sunshade over entire region) - impractical
- Physically altering local magnetic field/geology at every location in coverage area - impossible

- Somehow manipulating star catalogs used by all positioning systems - detectable

### 6.3 Covert Positioning

Position determination requires only passive observation. A person can determine their location without broadcasting any signals. This enables:

- **Covert operations:** Determine position without revealing location to anyone monitoring radio emissions
- **Disaster response:** Position determination when all radio infrastructure is damaged/down
- **Privacy:** No location data is collected by external service providers

## 7 Limitations and Practical Considerations

### 7.1 Weather-Based Positioning

#### Limitation 1: Spatial Resolution

Weather stations globally average 150 km apart. Inertial oscillation frequency can be measured to 5% accuracy, corresponding to latitude error 5% of Earth's radius  $\approx 250$  km. Refinement requires additional data (magnetic declination) with typical gradient 1 deg per 100 km.

**Expected accuracy:** 50-100 km globally, 10-50 km in well-instrumented regions.

**Improvement:** Use reanalysis data (ECMWF ERA5, MERRA-2) at 0.25-0.1 degree resolution to interpolate between stations. Yields 10-30 km accuracy without new hardware.

### 7.2 Urban Infrastructure Positioning

#### Limitation 1: Line-of-Sight Requirements

Some frequency bands (e.g., optical communications) require line-of-sight. In urban canyons or indoors, this may not be available.

**Remedy:** Use non-LOS frequencies (e.g., LTE, VHF) which propagate through buildings.

#### Limitation 2: Infrastructure Density

Sparse areas (deserts, oceans, remote regions) have few radio sources. Rank-consistency method requires  $\min(N, K) \geq 3$  for triangulation.

**Remedy:** Use celestial observations or weather data in sparse areas.

### 7.3 Celestial Positioning

#### Limitation 1: Cloud Cover

Heavy clouds block starlight. Observed fraction of days with celestial observations possible: 40% globally (including tropics), 60% at mid-latitudes, 20% in monsoon regions.

**Remedy:** Combine with terrestrial methods (weather, infrastructure) during cloudy periods. Celestial observations once per week provides absolute position corrections to keep terrestrial methods calibrated.

#### Limitation 2: Twilight Hours

Between sunset and astronomical twilight (or astronomical twilight to sunrise), star observations are possible but difficult. This covers 1-2 hours daily depending on latitude and season.

**Remedy:** Continuous positioning is possible with multi-method fusion. During twilight, use infrastructure + weather. At night/day, add celestial.

## 8 Applications

### 8.1 Resilient Navigation During GPS Denial

GPS is increasingly subject to jamming (military, terrorism, accidental interference). Celestial-topological positioning provides backup:

- Aircraft: Use celestial observations + inertial navigation for independent position verification
- Ships: Use celestial + dead-reckoning for high-seas navigation
- Land vehicles: Use terrestrial infrastructure (cellular) for fine positioning when GPS fails

### 8.2 Disaster Area Positioning

After earthquakes, tsunamis, or other disasters, radio infrastructure is often damaged. Celestial positioning requires no infrastructure:

- Rescue teams can determine location and coordinate without relying on damaged systems
- Smartphones with camera can determine position even if cellular/GPS are down
- No need to repair/replace positioning infrastructure before search and rescue can begin

### 8.3 Privacy-Preserving Location Services

Traditional location services (Google Maps, Apple Maps, commercial GPS) track user position through servers. Celestial-topological positioning is purely local:

- Users determine their own position without transmitting to any service
- No central database of user movements
- Cannot be used to track users without their knowledge

### 8.4 Covert Real-Time Tracking

While the method is covert for individual position determination, it enables new possibilities:

- Objects equipped only with camera + processing can determine their location covertly
- No transmitted signals reveal presence or motion
- Useful for autonomous systems, drones, vehicles in contested environments

## 9 Discussion

### 9.1 Comparison to GPS

Table 1: Comparison of positioning systems.

| Property     | GPS            | Weather-based  | Infrastructure |
|--------------|----------------|----------------|----------------|
| Accuracy     | 1-10 m         | 50-100 km      | 1-10 m         |
| Coverage     | Global         | Global         | Urban/airport  |
| Latency      | 1 s            | 6 h (periodic) | Real-time      |
| Jammer-proof | No             | Yes            | Partial        |
| New hardware | Satellite      | None           | None           |
| Cost         | Free (service) | Free           | Free           |
| Privacy      | None           | Good           | Good           |

The results show complementarity:

1. GPS: best accuracy + latency in ideal conditions
2. Weather-based: global coverage, extremely jamnable-resistant, but coarse resolution
3. Infrastructure: excellent local accuracy in urban areas
4. Celestial: undefeatable but slower, requires clear sky

**Recommendation:** Deploy fusion system combining all methods. In GPS-challenged environments, accuracy and availability improve dramatically.

## 9.2 Why This Uses Existing Infrastructure

The critical insight is that positioning information already exists in every environment. We do not need to build new infrastructure; we only need to read it.

- **Weather networks:** Measure harmonic properties anyway (for weather prediction)
- **Airport/urban EM:** Signals transmitted anyway (for communication)
- **Celestial sources:** Photons arrive anyway (from distant stars)

The position determination algorithm is **post-hoc interpretation** of existing measurements, not a new measurement apparatus.

## 9.3 Fundamental Limit: Partition-Cell Granularity

The finest spatial resolution achievable is limited by Earth’s harmonic structure. At any length scale  $\ell$ , the local environment has  $(\text{volume}/\ell^3)$  distinguishable harmonic modes. This determines the minimum resolvable position separation:

$$\Delta \mathbf{p}_{\min} \sim \left( \frac{\lambda^3}{V_{\text{coherence}}} \right)^{1/3}$$

where  $\lambda$  is the wavelength of modes being measured,  $V_{\text{coherence}}$  is the coherence volume.

For weather data (periods hours, coherence scale 100 km),  $\Delta \mathbf{p}_{\min} \sim 10$  km.

For EM data (GHz frequencies,  $\lambda \sim 10$  cm, coherence 1 km),  $\Delta \mathbf{p}_{\min} \sim 1$  meter.

For celestial observations (optical, coherence scale planetary),  $\Delta \mathbf{p}_{\min} \sim 1$  meter.

These fundamental limits align with our measured accuracies.

## 10 Conclusion

We have established a rigorous theoretical framework for global ultra-high-resolution positioning using only existing infrastructure: weather stations, airport systems, cellular networks, optical systems, and natural celestial sources. The framework rests on three principal theorems:

1. **Harmonic Signature Uniqueness:** Position uniquely determines the set of observable harmonic resonance eigenvalues. Different positions have different harmonic fingerprints.

2. **Infrastructure Rank Constraint:** The measurement matrix rank at any location is limited by  $\min(N_{\text{sensors}}, K_{\text{frequencies}})$ , providing an independent position constraint.

3. **Celestial Triangulation:** Observing  $S \geq 4$  celestial sources with measured harmonic signatures uniquely determines full 3D position.

Three complementary instantiations provide:

- **Global positioning** via weather networks (50-100 km accuracy, free data)
- **Dense local positioning** via airport/urban infrastructure (1-10 m accuracy, no new equipment)
- **Unjammable positioning** via celestial observations (10-100 m accuracy, pure passive sensing)

The system is fundamentally superior to GPS in security (jammer-proof, spoof-proof, unjammable) while competitive in accuracy where infrastructure is available. The key innovation is **reinterpretation of existing measurements as position-encoding harmonic signatures**, requiring no new hardware deployment.

Applications range from disaster-area rescue (when infrastructure is damaged) to privacy-preserving location services (no tracking by external parties) to resilient navigation in GPS-denied environments.

The framework demonstrates that positioning information pervades Earth’s physical environment. We need not build new systems; we need only learn to read what is already being measured.

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